BANGLADESH ARMY UNIVERSITY OF ENGINEERING & TECHNOLOGY (BAUET)

Department of Computer Science & Engineering

Software Development Project-I

NEXA AI

Your Computer, Finally Listening — in Your Language.

NEXA AI

A Local-First, Privacy-Preserving Desktop AI Assistant with Multilingual Voice Control & Policy-Gated Command Execution

TEAM MEMBERS

Supti Das Medha
0812310105101041

Md Abdur Rahman
0812220205101049

Shabbo Das
0812220205101096

SUPERVISOR

Md. Hashibur Rahman Kwoshik
Lecturer
Dept. of CSE, BAUET

1 PROBLEM STATEMENT

- Language exclusion limits AI access for Bangla speakers.
- Privacy risks from cloud-dependent assistants.
- Lack of local control over system commands.
- Unreliable outputs from rate-limited or offline services.

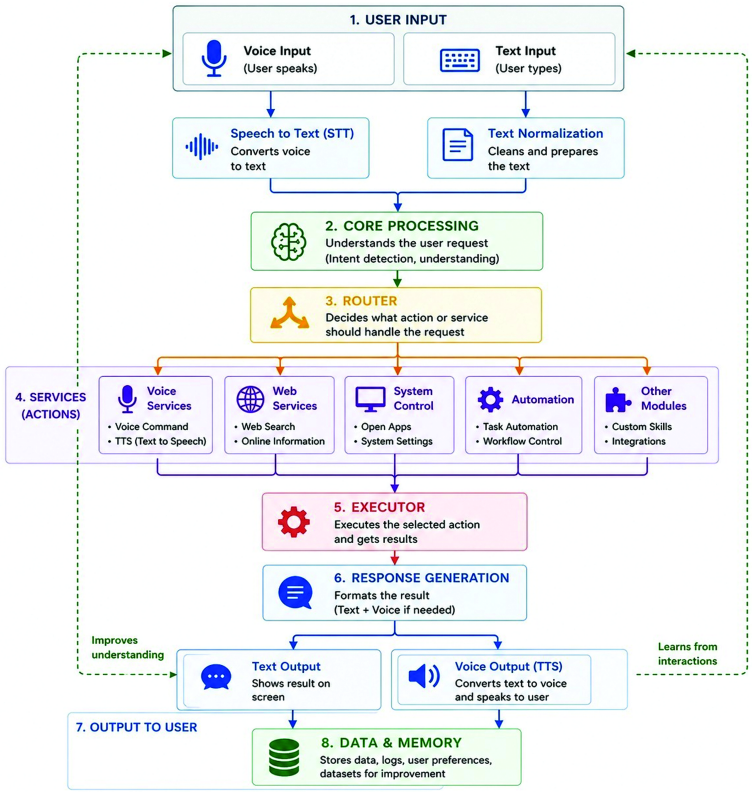
2 OBJECTIVES

- Provide Bangla, English & multilingual conversational AI.
- Ensure privacy with local-first processing.
- Execute whitelisted commands with user confirmation.
- Enable offline capabilities for critical operations.
- Maintain reliability and safety with a strict policy system.

3 OUR SOLUTION

Nexa AI is a local-first desktop assistant that understands Bangla, English and other languages. It processes most requests locally, ensures privacy, executes safe system actions only after user confirmation, and provides multilingual voice control for a seamless experience.

4 SYSTEM ARCHITECTURE



5 WORKFLOW (USER REQUEST TO RESPONSE)

6 KEY FEATURES

- Bangla, English & Multilingual Support
- Voice Recognition & Speech Response
- Local-First & Offline Capabilities
- Six-Provider LLM Failover System
- Whitelist-Based Command Execution
- Explicit Confirmation Before Action
- Real-Time Web Search & Knowledge Retrieval
- Audit Logging of All Activities
- Modern, Fast & User-Friendly UI

7 AI COMPONENTS

- **Intent Classifier** — Understand user requests
- **Policy Engine** — Enforce rules & safety
- **Tool Executor** — Execute system tools
- **Response Handler** — Generate & format replies
- **Voice Module** — STT, TTS & voice control

8 TECHNOLOGY STACK

Python · FastAPI · Electron · React · TypeScript · SQLite · Tailwind CSS · Vite · Redux

9 SECURITY & PRIVACY

- Local-first processing
- User data stays on device
- Whitelist-based execution
- Permission validation
- Audit logging & history
- No unauthorized actions
- Policy-gated command system

10 EXPECTED IMPACT

- Privacy-preserving AI experience
- Empower Bangla & local users
- Reduce dependency on cloud
- Increase productivity & efficiency
- Safer system automation
- Reliable offline functionality

11 FUTURE SCOPE

- Plugin marketplace
- Advanced RAG & vector DB
- Mobile companion app
- AI agent automation
- Cross-platform support
- Community tools & extensions

12 TARGET USERS

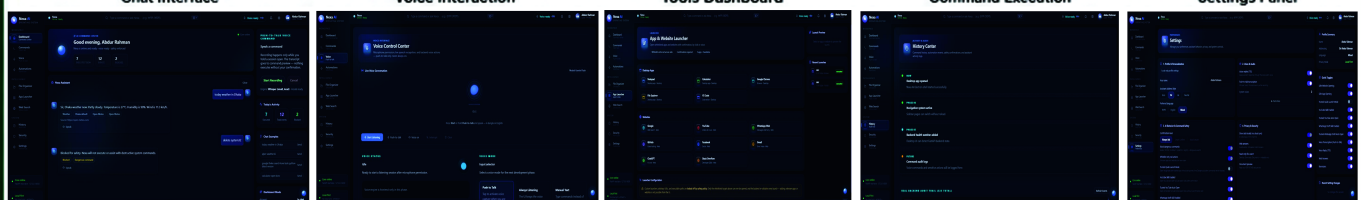
- Students & Researchers
- Developers & Engineers
- Professionals
- Bangla & Local Language Users
- Anyone needing a smart assistant

13 KEY METRICS (GOALS)

- 100% Local-First by Default
- < 2s Average Response Time
- 99% Uptime (Local Mode)
- Multi-Provider Reliability
- Zero Data Leak Policy
- High User Satisfaction

14 APPLICATION SCREENSHOTS (NEXA AI DESKTOP)

Chat Interface · Voice Interaction · Tools Dashboard · Command Execution · Settings Panel



15 TRY NEXA AI

Scan QR Code to watch demo video

Speaks Your Language. | Respects Your Privacy. | Asks Before It Acts. | Works Even Offline. | Built for Bangladesh.